Osptek Display

AMOLED SPECIFICATION

Model No:

AM173M466466FLS31

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REVISION RECORD

Version	Revision Items	Page	Name	Date
1.0	First release			20260314
2.0	TP IC 改为 CST9217			20260411
3.0	烧录亮度改为 800，更改大排走线			20260528

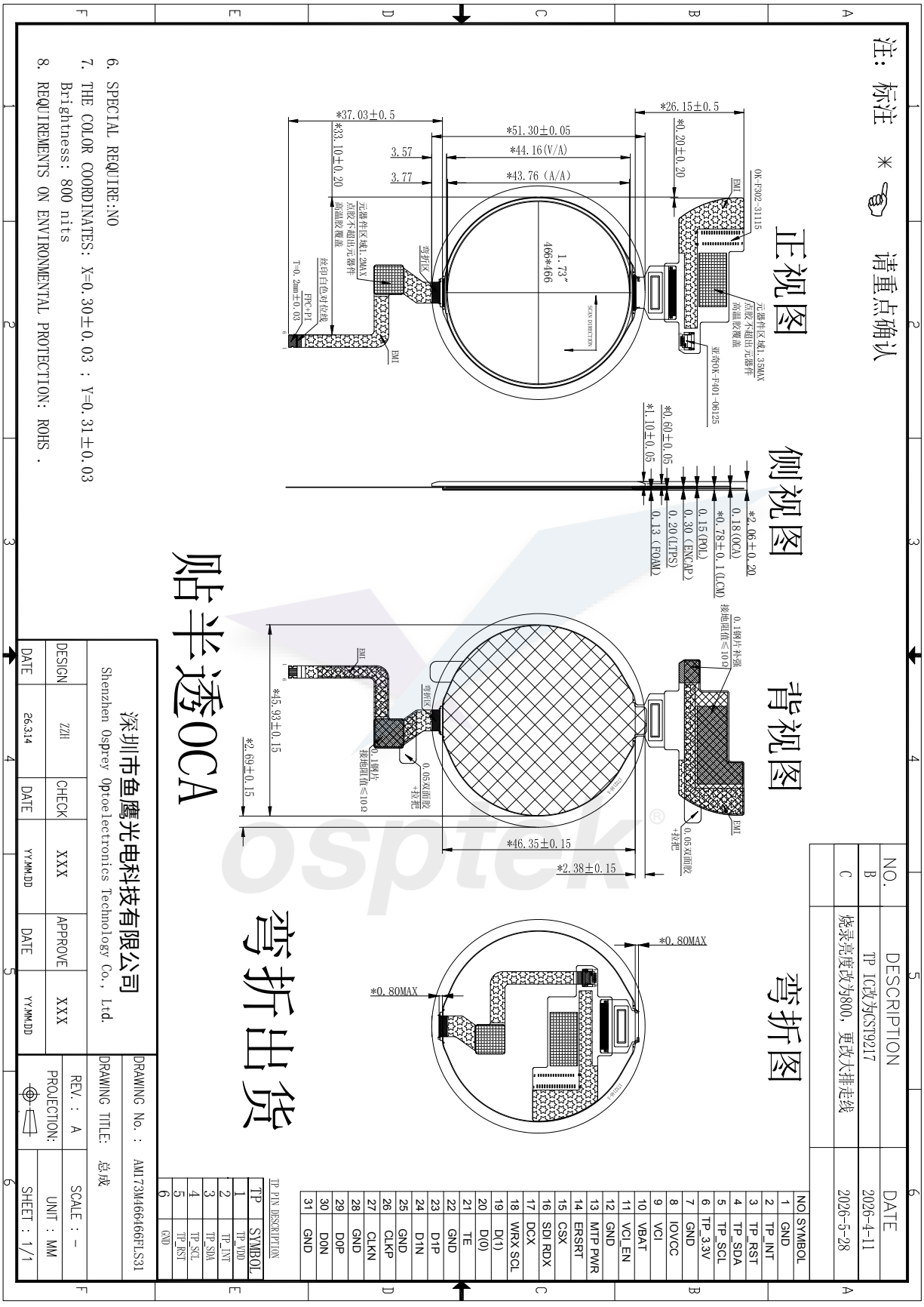
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1.General Spec

Item	Standard values	Unit
Diagonal Inch	1.73	inch
Dot arrangement	466(W) x RGB x 466(H)	dots
Pixel pitch	78 x 78	μm
Display mode	AMOLED	/
Driver IC | TP IC	CO5300 | CST9217	/
Module size	51.30(V) x51.30(W) x 2.06(T)	mm
Active area	Φ43.76	mm
Interface	QSPI/MIPI	-
Operating temperature	-10~60	℃
Storage temperature	-20~70	℃

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2.Mechanical drawing



3. Interface description

Pin No.	Symbol	Description
1	GND	Ground.
2	TPINT	Touch Panel interrupt
3	TPRST	Touch Panel Reset
4	TPSDA	Touch Panel Data
5	TPSCL	Touch Panel Clock
6	TPVDD	Touch Panel Power
7	GND	Ground.
8	VDDIO	Power supply for I/O block provided from outside VDDI < 0.05V (When power is turned off)
9	VDD	Power Supply for Analog Circuits VCI < 0.05V (When power is turned off)
10	VBAT	Analog Power Input for POWER IC Input Voltage Range: 3.4V to 5.5V
11	VCI_EN	Enable for POWER IC
12	GND	Ground.
13	MTP	External Voltage Input for OTP Data Program
14	RST	Device reset signal (0 : Enable ; 1: Disable)
15	CS	Chip Select Signal in MIPI DBI Type-B(MPU) and Type-C(SPI)
16	RDX	Serial input signal in QSPI I/F.
17	DCX	Display data / command selection in QSPI I/F.
18	SCL	A synchronous clock signal in QSPI I/F.
19	D1	Serial Data Input in Quad-SPI
20	D0	Serial Data Input in Quad-SPI
21	TE	Tearing effect output pin to synchronize MCU to frame Active Low
22	GND	Ground.
23	D1P	MIPI positive data signal
24	D1N	MIPI negative data signal
25	GND	Ground.
26	CLKP	MIPI positive clock signal
27	CLKN	MIPI negative clock signal
28	GND	Ground.
29	D0P	MIPI positive data signal
30	D0N	MIPI negative data signal
31	GND	Ground.

4. Absolute Maximum Ratings

Item	Symbol	Min	Max	Unit
Supply voltage (Display)	VDD	3.3	3.6	V
	IOVCC	1.8	3.6	V
	VBAT	3.4	5.5	V
	VCI_EN	3.3/1.8		V
Operating temperature range	Top	-10	+60	℃
Storage temperature range	Tst	-20	+70	℃
Storage humidity	HD	-	90	%RH

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5. DC Characteristics

DC Characteristics List

Parameter	Symbol	Conditions	MIN	TYP	MAX	Unit	Notes
Power generation & Operation Voltage							
AVDD booster voltage	AVDD	Operating Voltage	4.5	-	6.5	V	
VCL booster voltage	VCL	Operating Voltage	-5.0	-	-3.5	V	
Analog Operating voltage	VREFP	Operating Voltage	0.5	-	5.0	V	
Analog Operating voltage	VREFN	Operating Voltage	-5.0	-	-0.5	V	
Analog Operating voltage	I_ELVDD	Operating Voltage	2.0	-	5.0	V	
Analog Operating voltage	I_ELVSS	Operating Voltage	-5.4	-	-0.5	V	
Gamma reference voltage	VGMP	Operating Voltage	2	-	6.3	V	
Gamma reference voltage	VGSP	Operating Voltage	0	-	4.5	V	
VGH booster voltage	VGH	Operating Voltage	3	-	10.6	V	
VGL booster voltage	VGL	Operating Voltage	-15	-	-2	V	
Voltage difference between VGH and VGL	VGH-VGL	|VGH-VGL| ≦30			30	V	
I/O operating voltage	VDDI	I/O supply voltage	1.65	-	3.3	V	
LOGIC INPUT/ OUTPUT							
Logic High level input voltage	V_{IH}		0.8× VDDI	-	VDDI	V	1
Logic Low level input voltage	V_{IL}	-	VSS	-	0.2× VDDI	V	1
Logic High level output voltage	V_{OH}	I_{OH} = -0.1mA	0.8× VDDI	-	VDDI	V	2
Logic Low level output voltage	V_{OL}	IOL = +0.1mA	VSS	-	0.2× VDDI	V	2
Logic High level leakage	I_{LIH1}	Vin = 0 to *VDDI*	-	-	1	μA	1,2
Logic Low level leakage	I_{LIL1}	Vin = 0 to *VDDI*	-1	-	-	μA	1,2
Source OP Output							
Output deviation voltage	V_{dev}	Sout ≧ 4.2V Sout ≦ 0.8V			TBD	mV	4
Output deviation voltage	V_{dev}	4.2V>Sout>0.8V			TBD	mV	
Output offset voltage	V_{OFSET}				TBD	mv	
Stand-by Current							
Sleep In mode	I_{stlp1}	DSI LP mode VDDI Current		TBD		μA	
		DSI LP mode VCI Current		TBD		μA	1
	I_{stu1}	DSI Ultra Low power VDDI Current		TBD		μA	
		DSI Ultra Low power VCI Current		TBD		μA	1
Oscillator Output							
Oscillator tolerance	△OSC	All Temperature	-2%	-	2%	%	3

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6.Electron-optical Characteristics

Item	Symbol	Min.	Typ.	Max.	Unit	Remark
AMOLED Power positive	ELVDD				V	Ref
AMOLED Power negative	ELVSS				V	Ref
Digital Power supply	VDDIO	1.7	1.8	3.3	V	Ref
Analog Power supply	VCI	2.7	2.8	2.9	V	Ref

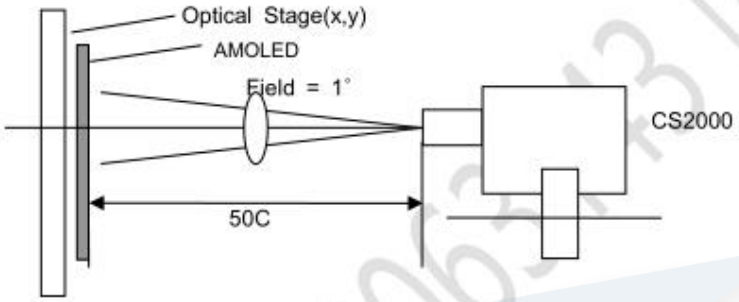
Item		Symbol	Condition	Value			Unit	Note
				Min	Typ	Max		
Luminance		Bp		500			nit	Normal mode with CG
Uniformity		△Bp	W255	80	85		%	Note 6
Viewing Angle	Left	θ_L	CR≥10			80	°	Note 2
	Right	θ_R				80		
	Top	ψ_T				80		
	Bottom	ψ_B				80		
Contrast Ratio		Cr	Θ=0°	100000:1				Note 3
Color Coordinate of CIE1931 (with lens)	Red	X						
		Y						
	Green	X						
		Y						
	Blue	X						
		Y						
	White	X		0.280	0.300	0.320		
		Y		0.290	0.310	0.330		
NTSC Ratio		NTSC		97	105		%	Note 4

Flicker	amount	60Hz, line interleaved pattern			-50	db	Note 5
Crosstalk	△CT				2	%	Note 8
Gamma	-		2.0	2.2	2.4		Normal mode

*** Reference value.**

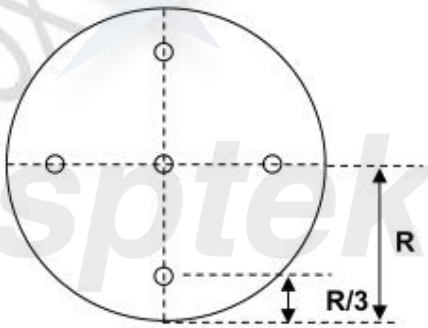
1. the ambient temperature is 25°C.po
2. The test systems refer to Note1 and Note2.

Note 1: Definition of optical measurement system.



Optical Characteristic Measurement Equipment and Method

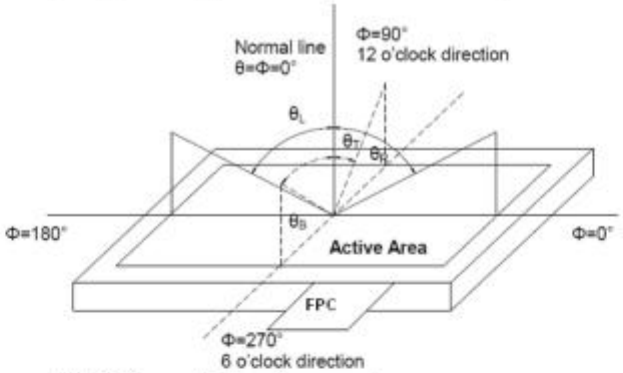
The optical characteristics should be measured in dark room. The optical properties are measured at the center point of the AMOLED screen. All input terminals AMOLED panel must be ground when measuring the center area of the panel.



Measuring point for surface luminance (5points)

Visionox V0063-13 内部公开
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Note 2: Definition of viewing angle range and measurement system.



Definition of viewing angle

Note 3: Definition of contrast ratio

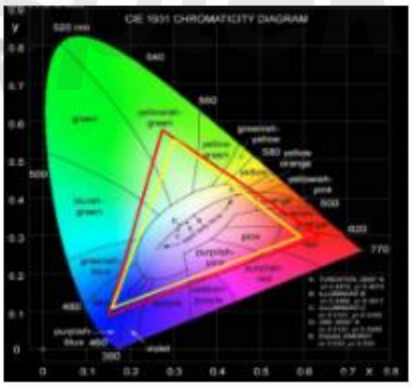
$$\textbf{Contrast ratio(CR)} = \frac{\textbf{Luminance measured when AMOLED is on the "white" state}}{\textbf{Luminance measured when AMOLED is on the "Black" state}}$$

"White state ": A state where the AMOLED should be driven by Vwhite.

"Black state": A state where the AMOLED should be driven by Vblack.

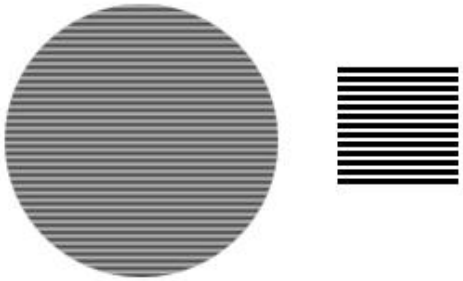
Note 4: Definition of color chromaticity (CIE1931)

Color coordinates measured at center point of AMOLED.



Note 5: Flicker

Adjust the sample to W255 and line interleaved pattern, and measure Flicker value by JEITA with CA310.



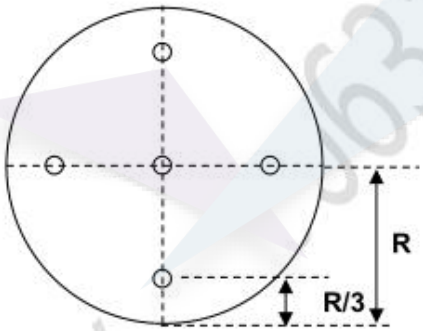
隔行亮暗 pattern

Note 6: Definition of luminance uniformity

5 measuring points is located at the Active area as shown below.

Luminance Uniformity(U) = Lmin/ Lmax

L-------Active area length W----- Active area width

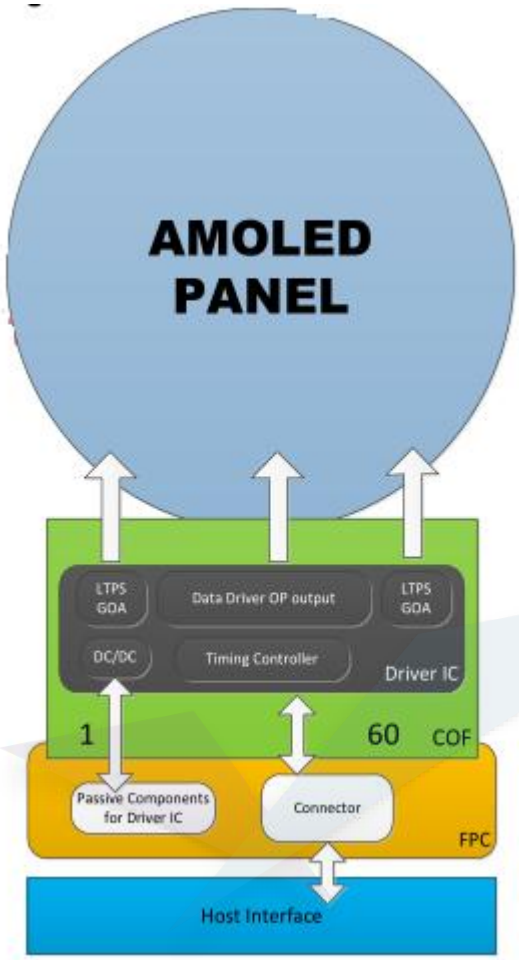


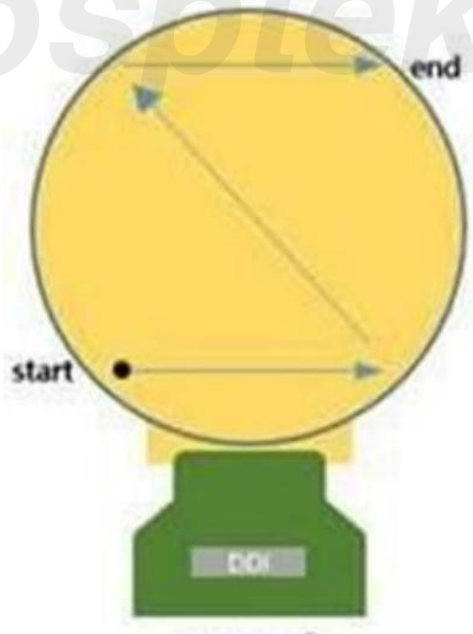
Definition of uniformity (5points)

Lmax: The measured maximum luminance of all measurement position.

Lmin: The measured minimum luminance of all measurement position.

7. System Block Diagram





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8. AC characteristic& Power On/Off Sequence

QSPI write protocol

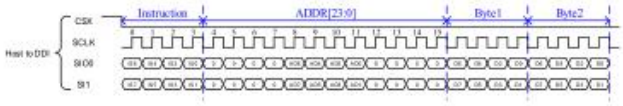
- QSPI Command/pixel write:支持888/666/565/332/111/GRAY256 data format

1 wire write



666/565/332格式支持补位方式：
补MSB/补0/补1

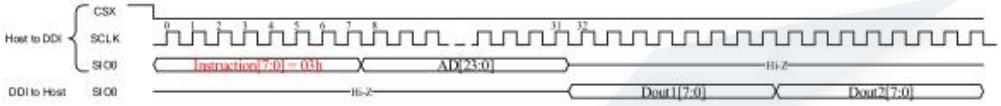
2 wire write



4 wire write

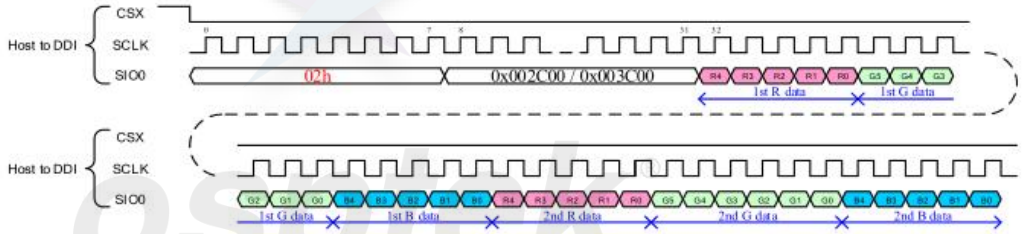


- QSPI Command/pixel read

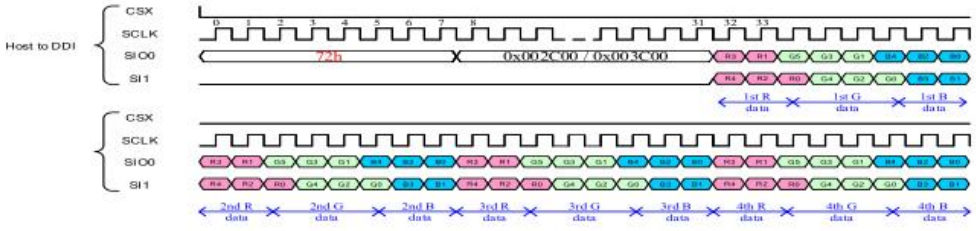


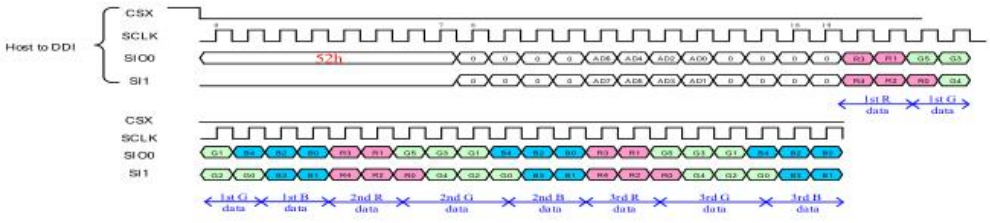
QSPI Data format for RGB565

1 wire



2 wire

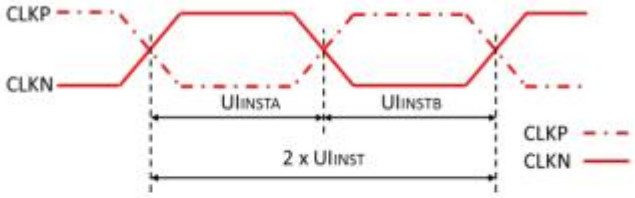




9.High speed clock and data timing

MIPI AC Characteristics

6.5.1 High Speed Mode - Clock Timings



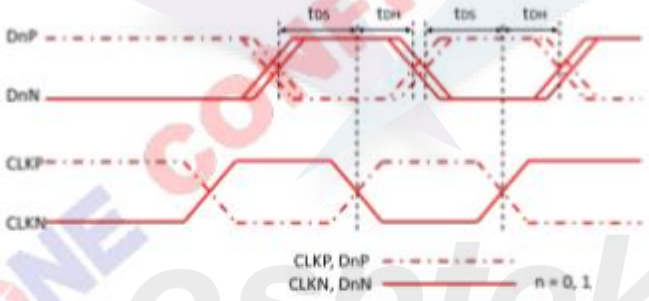
Clock Timing

High Speed Mode - Clock Timing

Signal	Symbol	Parameter	Specification			Unit	Notes
			MIN	TYP	MAX		
CLK P/N	2xUIINST	Double UI instantaneous	2		25	nS	
CLK P/N	UIINSTA, UIINSTB	UI instantaneous Half	4		12.5	nS	1

Note 1: *UI = UIINSTA = UIINSTB.*

6.5.2 High Speed Mode - Clock / Data Timings



DSI Clock / Data Timings

High Speed Mode - Clock / Data Timing

Signal	Symbol	Parameter	Specification			Unit	Notes
			MIN	TYP	MAX		
Dn P/N	tDS	Data to Clock Setup time	0.15*UI			UI	
(n=0, and1)	tDH	Clock to Data Hold time	0.15*UI			UI	

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High Speed Mode - Rising and Falling Timings

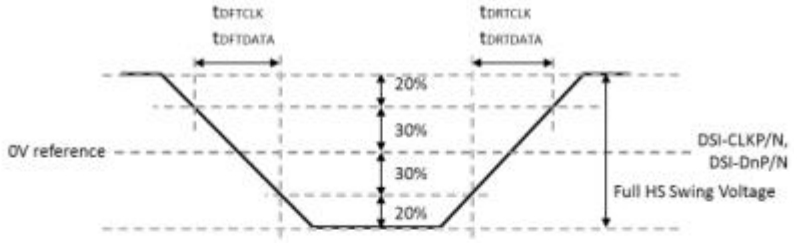


Figure 6-2 Rising and Falling Timings

High Speed Mode - Rising and Falling Timing

Parameter	Symbol	Conditions	Specification			Unit	Notes
			MIN	TYP	MAX		
Differential Rise Time for Clock	t_{DRTCLK}	CLKP/N	150pS		0.3*UI		2,3
Differential Rise Time for Data	$t_{DRTDATA}$	DnP/N	150pS		0.3*UI		1,2,3
Differential Fall Time for Clock	t_{DFTCLK}	CLKP/N	150pS		0.3*UI		2,3
Differential Fall Time for Data	$t_{DFTDATA}$	DnP/N	150pS		0.3*UI		1,2,3

Note 1: *DnP/N, n =0, and 1.*

Note 2: *The display module has to meet timing requirements, which are defined for the transmitter (MCU) on MIPI D-PHY standard.*

Note 3: *DSI-CLK+ = CLKP.*
DSI-CLK- =CLKN.
DSI-D0+ =D0P.
DSI-D0- =D0N.

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10.Reliability test conditions and methods

No	Test Item	Test condition	Criterion
1	High Temperature Storage	70℃±3℃ 48H Power off	Inspection after 4 hours storage at room temperature,the sample shall be free from defects: 1. Air bubble in the LCD; 2.Seal leak; 3.Non-display; 4.missing segments; 5.Glass crack; 6.Current Idd is twice higher than initial value. 7.The surface shall be free from damage. 8.The electric characteristic requirements shall be satisfied
2	Low Temperature Storage	-20℃±3℃ 48H Power off	
3	High Temperature Operation	60℃±3℃ 48H Power on	
4	Low Temperature Operation	-10℃±3℃ 48H Power on	
5	High Temperature & Humidity Operation	60℃±3℃ 90%RH 48H Power off	
6	Temperature Cycle	-20℃←→25℃←→70℃ 30min 5min 30min 5 cycle Power off	

Remark:

1.The test samples should be applied to only one test item.
2.Sample size for each test item is 3~5pcs.
3.For Damp Proof Test, Pure water(Resistance＞10MΩ) should be used.
4.In case of malfunction defect caused by ESD damage, if it would be recovered to normal state after resetting, it would be judged as a good part.
5.EL evaluation should be excepted from reliability test with humidity and temperature: Some defects such as black spot/blemish can happen
by natural chemical reaction with humidity and Fluorescence EL has.
6.Failure Judgment Criterion: Basic Specification, Electrical Characteristic, Mechanical Characteristic, Optical Characteristic.